



Sumer Iqbal

Researches privacy-preserving federated learning for brain tumour segmentation.



1 project · 1 research paper · part of AI Collective

RESEARCH

All research is carried out under the supervision of **Dr. Umme Zahoora**. Results are from the manuscripts and may change in review.

● READY TO SUBMIT · FRONTIERS IN ONCOLOGY **FIRST AUTHOR**

Federated learning · Medical imaging

A Privacy-Preserving Federated Learning Framework with Variance-Based Aggregation and Customized U-Net for Brain Tumor Segmentation

Sumer Iqbal, Khadija Tul Kubra, **Muhammad Azeem Bhatti**, Farquleet Farhat Gondal, Saddam Hussain Khan, Umme Zahoora

Trains a lightweight attention U-Net across hospitals without moving any scans. A new aggregation method, Fed-PLM, merges each site's update along its direction of greatest variation; on BraTS 2021 the federated model comes within 0.01 Dice of centralised training.

WHOLE-TUMOUR DICE

0.885 federated

CENTRALISED BOUND

0.894

SCANS SHARED

None

Federated learning

Fed-PLM

Attention U-Net

BraTS2021

PROJECTS

CONTRIBUTOR

 Source

Computer vision · PyTorch · Jan 2026

GAN-balanced plant disease classification [↗](#)

Does generating minority-class images fix an imbalanced dataset?

A GAN generates extra leaf images for the under-represented classes of a 38-class plant disease dataset. A custom CNN is then trained on the balanced set and checked with 5-fold cross-validation.

ACCURACY

98.19%

MACRO F1

0.982

CLASSES

38

GAN

CNN

Class imbalance

K-fold

[Dataset on Kaggle](#) [↗](#)